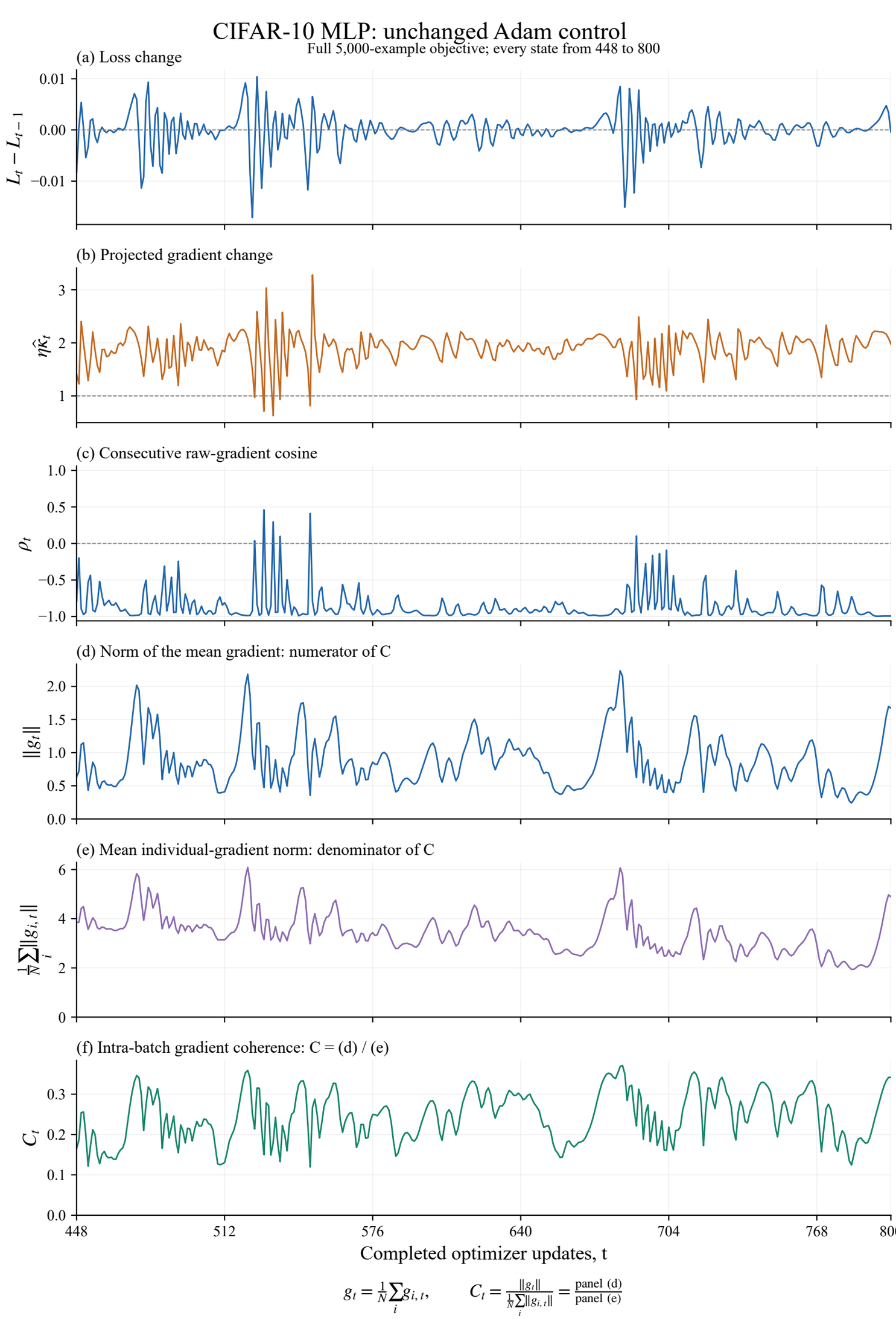

CIFAR-10 MLP: unchanged Adam control
Full 5,000-example objective; every state from 448 to 800
(a) Loss change
$L_t - L_{t-1}$
0.01
0.00
−0.01
(b) Projected gradient change
$\hat{\eta}\hat{\kappa}_t$
3
2
1
(c) Consecutive raw-gradient cosine
ρ_t
1.0
0.5
0.0
−0.5
−1.0
(d) Norm of the mean gradient: numerator of C
$\|g_t\|$
2.0
1.5
1.0
0.5
0.0
(e) Mean individual-gradient norm: denominator of C
$\frac{1}{N}\sum_i \|g_{i,t}\|$
6
4
2
0
(f) Intra-batch gradient coherence: C = (d) / (e)
C_t
0.3
0.2
0.1
0.0
448
512
576
640
704
768
800
Completed optimizer updates, t
$g_t = \frac{1}{N}\sum_i g_{i,t}$,
$C_t = \frac{\|g_t\|}{\frac{1}{N}\sum_i \|g_{i,t}\|} = \frac{\text{panel (d)}}{\text{panel (e)}}$
Raw gradients include all weights and biases. No smoothing or subsampling.